Prodigy Anesthesia Quick Review

Antacids

What is the principal indications for antacids prior to surgery?	They are administered to neutralize gastric acid that is present in the stomach. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 603.*
Which are preferred for raising the gastric pH, particulate antacids or nonparticulate antacids and why?	Nonparticulate antacids such as sodium citrate are preferred over particulate antacids such as Maalox or Mylanta because the particles can cause pulmonary injury if aspirated. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 941.*
How effective are antacids at raising the gastric pH?	An antacid dose administered 15-30 minutes prior to induction is 100% effective at raising the pH of gastric fluid above 2.5. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 603.*
How does the administration of antacids affect drug absorption?	Antacids can slow the absorption of PO digoxin, cimetidine, and ranitidine, but speed the absorption of phenobarbital. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 281-282.*
How do antacids affect gastric emptying time?	Antacids slow gastric emptying and increase gastric volume *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 282.*
The administration of antacids obviously increases the gastric fluid volume. Is this a concern with regard to increased risk of aspiration?	The risk of aspiration depends upon both the pH of the fluid in the stomach as well as the volume. The fact that antacids increase the volume of fluid in the stomach, however, does not warrant witholding antacids. It has been shown that the mortality rate of aspiration is increased when small volumes (0.3 mL/kg) of acidic fluid is aspirated compared to larger volumes of buffered solution. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 603.*
Is the administration of an antacid useful in an emergency setting?	Yes. In fact, because it has an almost immediate and short-lived peak effect (15-20 minutes), it is more useful as an adjunct to emergency procedures rather than outpatient procedures where the time between administration of the antacid and induction of anesthesia may be prolonged. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 940-941.*

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What is the pediatric dose of sodium citrate?	0.4 mL/kg of 0.3 M sodium citrate *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 941.*
What is the adult dose of sodium citrate?	30 mL *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 941.*
What are two examples of nonparticulate antacids?	Sodium citrate (Bicitra) and two Alka-Seltzer Gold tablets (which contains both sodium citrate and potassium citrate) dissolved in 30 mL of water are both options for nonparticulate antacids. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 941.*

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